

Time to Leave the Louvre: A Computational Network Analysis

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Summary

Increase in terror attacks has raised demand for safe emergency evacuation plans worldwide. We focus on evacuating the Louvre, the world's largest art museum. Evacuation is made difficult by the volume and variety of visitors; the Louvre management desires evacuation plans over a broad set of considerations.

We partition the Louvre into sections and build an agent-based model to simulate evacuations in each section. We run simulations over each section to determine a rate by which agents exited. To connect sections, we represent the building as a graph, thereby posing a network flow problem. The strong duality property identifies bottleneck edges in the graph. Simulating blocked passages or new secret exits is simply removal or addition of edges to the graph. Bottleneck identification is our highest priority for public safety.

Our model predicts that an evacuation plan using all four public exits could evacuate the Louvre in 24 minutes. Furthermore, while many bottlenecks surround the Pyramid entrance, the entrance itself is not a bottleneck. This property of the Pyramid is crucial in emergencies, since it allows access for emergency personnel. Additionally, securing the Passage Richelieu is critical to evacuation, since its safety is directly linked to the Pyramid's safety. Keeping these entrances open and useful is imperative to both speed and safety of an evacuation.

Our model is powerful due to its ability to model individual human behavior, followed by an adaptable abstraction of building-flow dynamics. One weakness of our model is that we consider worst-case scenarios, but the evacuation times are an upper bound for a real evacuation.

Restatement of the Problem

We are tasked with the broad problem of designing an evacuation model for the Louvre that allows exploration of a range of options. Our primary goals are to:

- determine a means to assess the efficiency of a given evacuation plan,
- develop an optimal evacuation plan without compromising safety,
- identify key bottlenecks and other obstacles to safe and efficient evacuation,
- determine the effect of additional exits or blocked routes on the evacuation, and
- communicate a clear plan of implementation.

Secondarily, we are also to consider:

- the effect of the diversity of visitors (language, grouping, disability) on evacuation;
- potential benefits of technology in aiding evacuation;
- deployment routes for emergency personnel; and
- adaptability of the model to other buildings.

General Assumptions

- **Self-interest.** Evacuees will not consider a globally optimal solution for everyone but will instead choose a locally “greedy” solution.
- **Natural flow.** Individuals leave by the closest exit, unless directed otherwise.
- **Adherence to procedure.** Individuals follow the evacuation plan provided by Louvre management.
- **Safety outside.** Securing the safety of people once outside of the building is outside of the scope of our task.
- **Importance of panic.** Increasing panic causes suboptimal or irrational choices.
- **Elevators used only by emergency personnel and disabled people.** Elevators can be dangerous in emergency situations.
- **Appropriate signage and technology in multiple languages.** Much of the Louvre’s signage contains universally-comprehensible symbolic instructions. Moreover, software packages and phone apps giving evacuation directions are easily written for different languages.

Introduction: Definitions and Roadmap

An *evacuation plan* is a collection of pathing procedures for evacuees, conditional on the location and state of each evacuee. The simplest evacuation plan would adhere to “natural flow,” in which evacuees move towards the nearest exit.

To measure the efficiency of each evacuation plan, we could estimate the time to fully empty the building. However, this measure is somewhat naïve, since it discounts safety and is highly dependent on initial conditions. For example, with unusually many visitors clustered around the Mona Lisa, attempting to funnel all of them through one exit could result in overcrowding, trampling, and mob panic risks.

As an alternative, we consider the *maximum exit rate* or, mathematically, the maximum of the time derivative of exiting evacuees. This would seem to be an identical measure, since a decrease in total time would mean an increase in average exit rate. However, the exit rate reaches a peak sometime during the middle of evacuation, as in **Figure 1**.

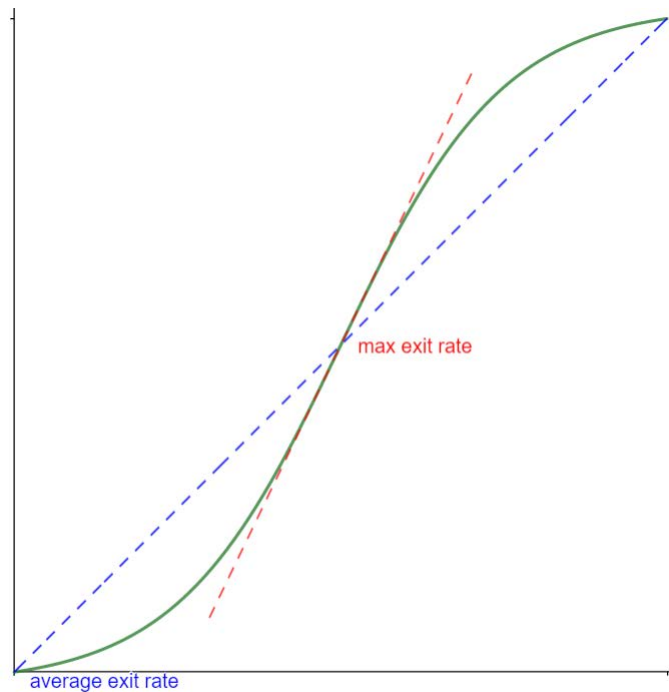


Figure 1. Exited evacuees vs. time elapsed.

Optimizing for maximum exit rate optimizes the *throughput* of evacuees through the Louvre rather than the minimum total time to evacuate. The value is twofold:

- If the Louvre is at high occupancy, average exit rates should approach maximum exit rates.
- If the Louvre is at low occupancy, higher throughput should decrease

crowding risks.

As a result, maximizing throughput is directly correlated with maximizing public safety.

Modeling Roadmap

We implement a two-stage model:

- The first stage uses computational agent-based modeling to understand local evacuee flow dynamics within sections of the Louvre.
- The second stage combines information from the sections into a flow network for which we can assess and optimize evacuation plans.

To clarify, a roadmap for our model is:

- Partition each floor of the Louvre into smaller subsections.
- Develop a computational agent-based model to study local evacuation phenomena and evacuee flow for each subsection.
- Develop a global network that models each subsection as a node, passageways as edges, and evacuee flow as weights.
- Apply graph algorithms to maximize evacuee flow and predict the effect of adding or removing edges.
- Interpret results into real-world terms and useful policy suggestions.

Part I: The Local Section Model

Partitioning Sections

The Louvre consists of nested gallery layouts, several access points to other floors, and multiple exit points [Louvre Museum 2016]. By partitioning the museum into smaller, less-complex subsections to model individually, we reduce complexity.

We split each floor into the five subsections (**Figure 2**) and label the sections A to E (for example, the bottom-left section on the ground floor is labeled “ground floor A”). The Napoleon Hall, denoted “Napoleon P,” has the Pyramid entrance (as shown by **Figure 3**), which is the only relevant subsection on that floor.

Development

We develop our local evacuation model (the Local Section Model) in NetLogo, agent-based modeling software. The main idea of agent-based

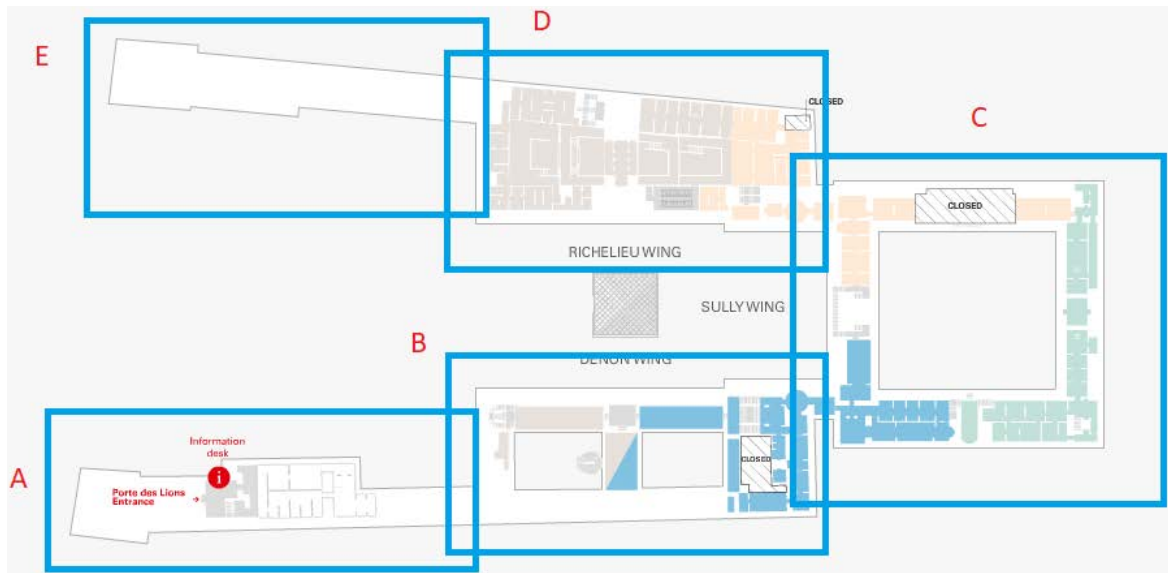


Figure 2. Each floor of the Louvre is split into five subsections, labeled A to E, starting from the bottom and moving counter-clockwise.

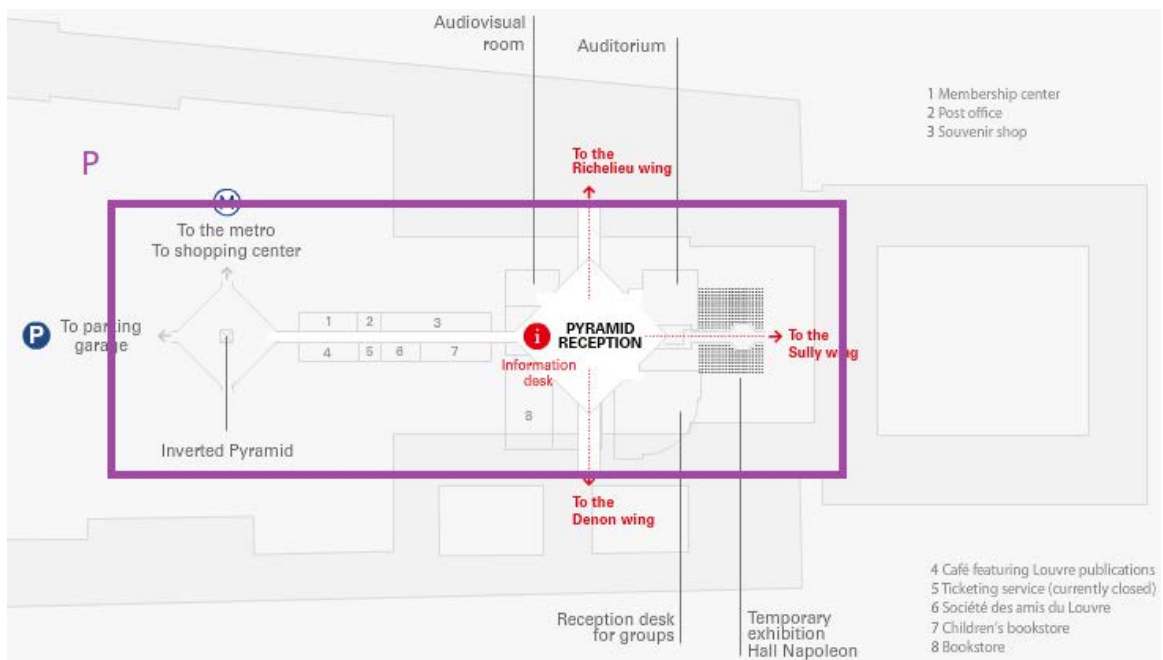


Figure 3. The Napoleon Hall contains a Pyramidal subsection unique to its floor.

modeling is that agents are single units with specific, well-defined goals. While an individual agent's behavior is typically simple, the complex behavior of a system of agents is usually more than the sum of its parts. Since each person acts solely in their own self-interest to evacuate, an agent-based model is highly applicable.

Figure 4 shows the resultant interface in NetLogo, which is easy to use and interpret. The bottom of the figure shows section ground floor D with its complex gallery system. The white agents acting on a grid of green (passable) and black (impassable) patches represent evacuees finding their way through galleries. Blue patches (such as the vertical line segment at extreme upper right) represent entryways through which evacuees enter, and red patches (such as the vertical line segment at extreme upper left) represent exits through which evacuees leave.

The specific behavior logic of the agents is shown in **Figure 5**. We equip each agent with a variable panic attribute and fixed speed attribute. Further information on the speed attribute is contained in the following section, while the panic attribute is explained here.

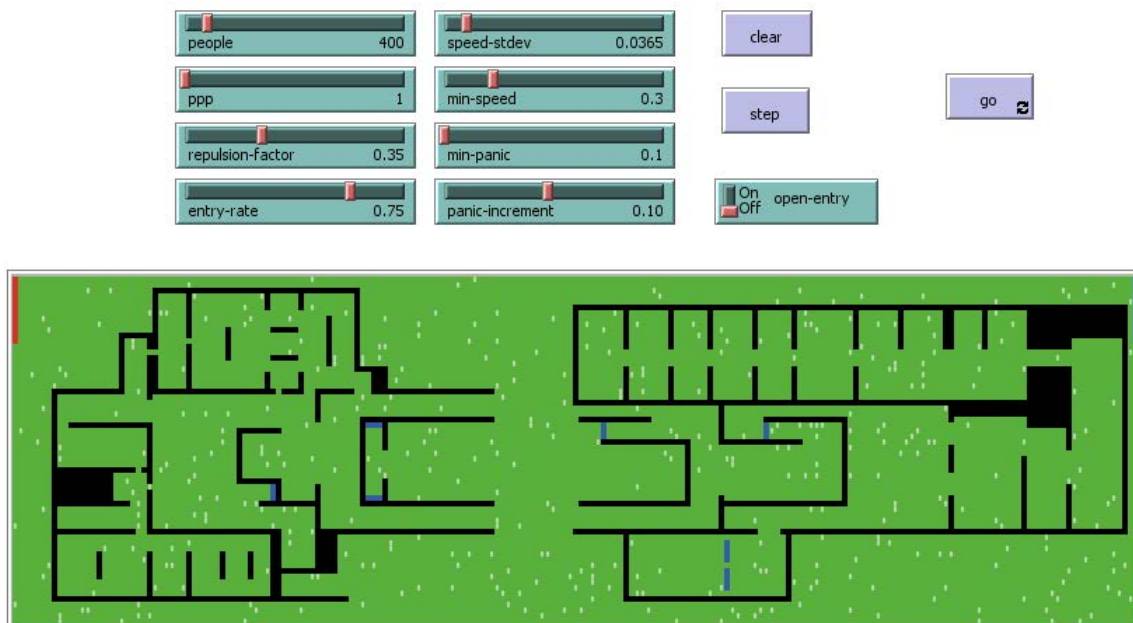


Figure 4. Representation of the galleries on ground D.

The logic in **Figure 5** is consistent with our assumption that each agent acts in their own self-interest, such as making locally optimal steps towards the nearest exit. However, this factor alone causes purely deterministic behavior that does not appear to model human movements accurately, especially in a tense situation. To accommodate such behavior, we add a panic parameter to represent increased tension in the individual, as evacuees react to being unable to move (due to crowd density) by shuffling around, looking for other exit routes, and so forth. While it is difficult to quantify panic, it is not difficult to include its effects in a model. We take inspira-

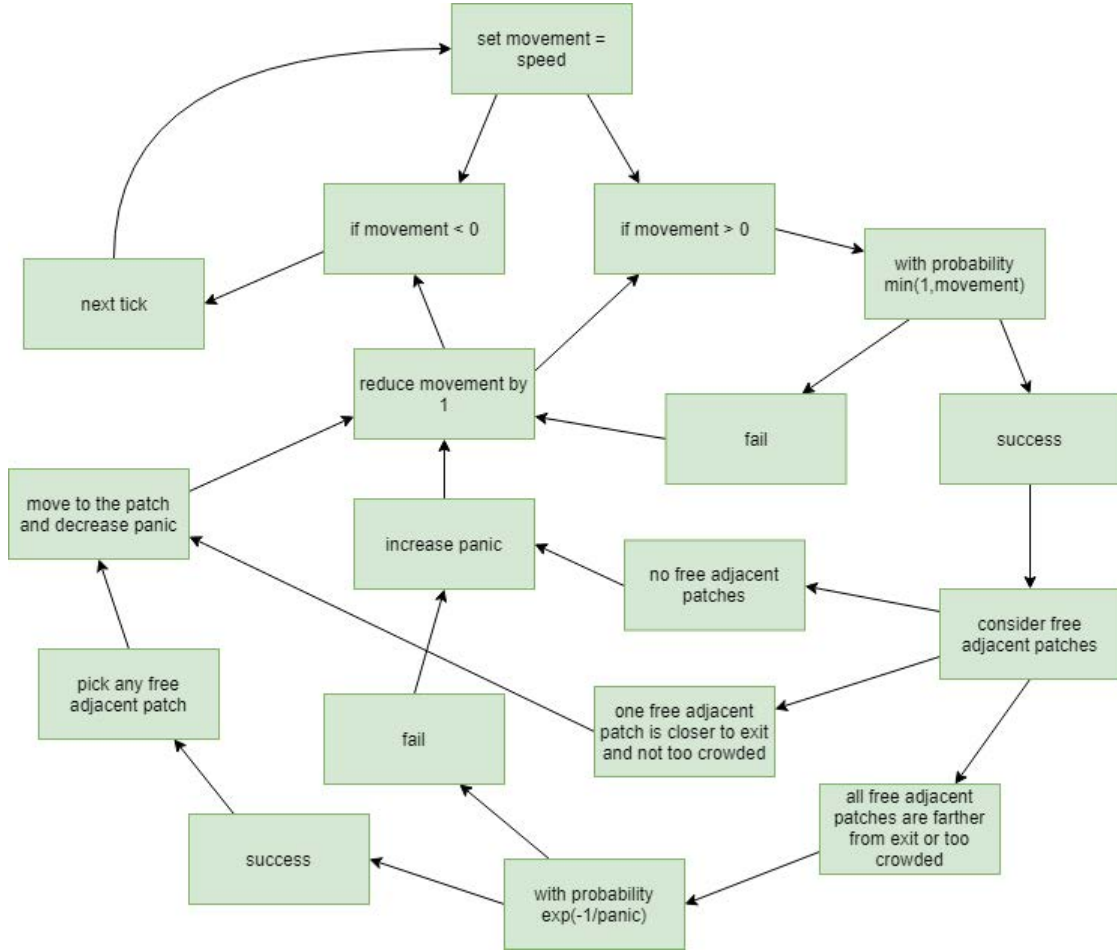


Figure 5. Flowchart of an individual agent's behavior logic.

tion from simulated annealing [Wikipedia 2019a]. We increase the agent's panic each time step that it is stationary and compute p_m , the probability of moving slightly farther from their destination, by

$$p_m = \exp\left(\frac{-1}{\text{panic}}\right),$$

with

$$\lim_{\text{panic} \rightarrow 0^+} p_m = 0, \quad \lim_{\text{panic} \rightarrow \infty} p_m = 1.$$

The top of **Figure 4** shows slider parameters:

- people indicates the initial population of agents in the room.
- ppp indicates the maximum density of agents per passable patch. Scaling this parameter up increases the maximum allowable crowdedness.
- repulsion-factor is related to what each agent determines to be “too crowded” during local pathfinding. Scaling this parameter up decreases crowding tolerance.

- `speed-stdev` refers to the variance of speed attributes in the population of agents.
- `min-speed` gives the minimum speed of the agent.
- `panic-increment` refers to how much more panicked each agent gets when unable to move.
- `min-panic` gives the minimum panic attribute of the agent.
- the slider `entry-rate` and the switch `open-entry` are related to spawning new agents through the blue entry patches.

Physical Interpretability

To scale our model to the dimensions of the Louvre, we use Google Maps. **Figure 6** shows points that we use as reference and examples of calculating distances via the Python package GeoPy.

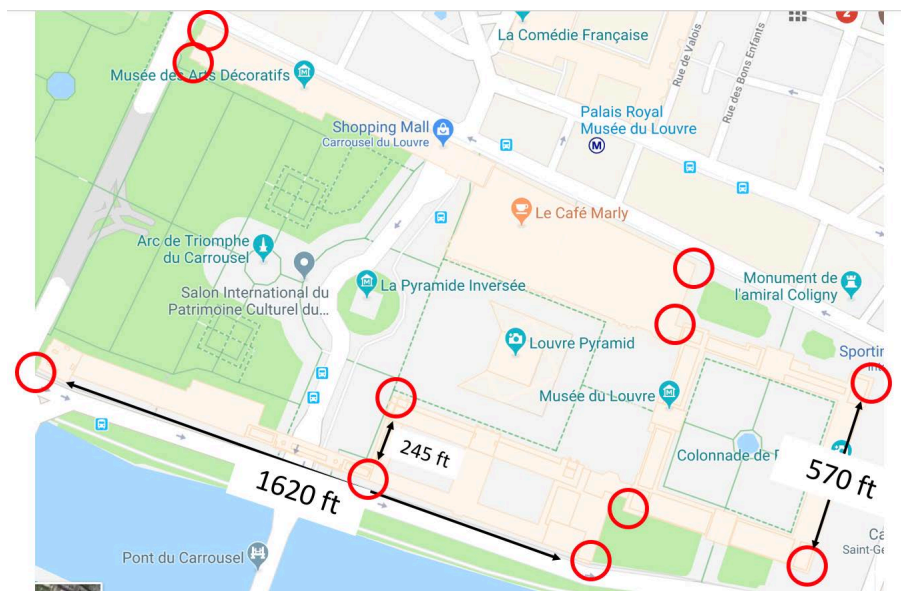


Figure 6. Points pinged on Google Maps, with some example distances.

To determine the speeds of agents, we refer to Yosritzal et al. [2018], who simulated a tsunami evacuation in Indonesia. Their results give an average walking speed of an individual during an evacuation as $1.419 \text{ m/s} \approx 4.656 \text{ m/s}$. Hence, we scale each subsection's representation so that one *patch* is 4 ft by 4 ft. As a result, one *tick* (the unit time in Netlogo) in the model's progression is roughly equivalent to one second in real-world time. Since our agents can move in one tick in either a cardinal direction (4 ft) or a diagonal direction ($4\sqrt{2} \approx 5.66 \text{ ft}$), it is reasonable to translate 4.656 ft/s into 1 patch per tick.

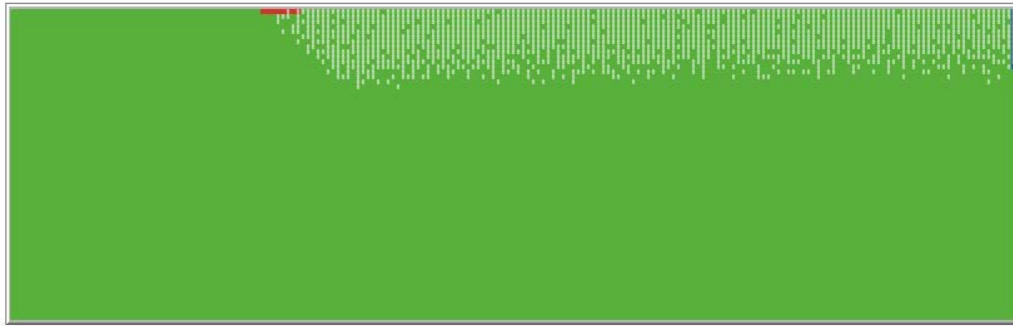
Additionally, to model the varying walking speeds of a diverse population, we approximate the distribution of walking speeds with a normal

distribution with mean 4.656 ft/s and standard deviation 0.170 ft/s. This distribution was derived from the statistics of Yosritzal et al. [2018] on a population of ages 20–60.

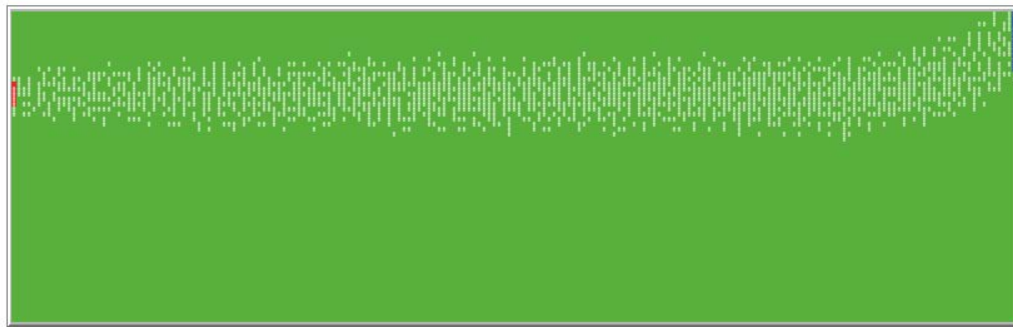
Corner Bottlenecking: The Price of Turning

Turning and the presence of corners present bottlenecks to optimal evacuee flow in the Local Section Model.

Given an individual agent's wish to take the shortest path to an exit, compression arises at both corners and orthogonal exits (an exit perpendicular to the entry corridor from which the agent entered). **Figure 7a** gives an example of this phenomenon over the Porte des Lions exit of ground floor A, where we can clearly see the effect of the individual agent's desire to escape as quickly as possible: a triangular wedge formation that restricts maximum flow outwards, rather than the linear flow described by **Figure 7b**. While mitigated significantly by our agents' repulsion force, the effect is still significant: The orthogonal exit modeled in **Figure 7a** has a maximum outflow of 4.2 e/s (evacuees per second), the linear exit modeled in **Figure 7b** has a maximum outflow of 4.8 agents/s, a 14% increase.



a. The effect on exit flow of turning: 4.2 e/s.

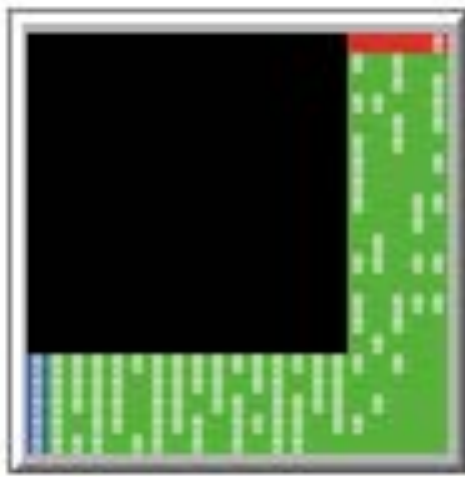


b. Linear exit flow: 4.8 e/s.

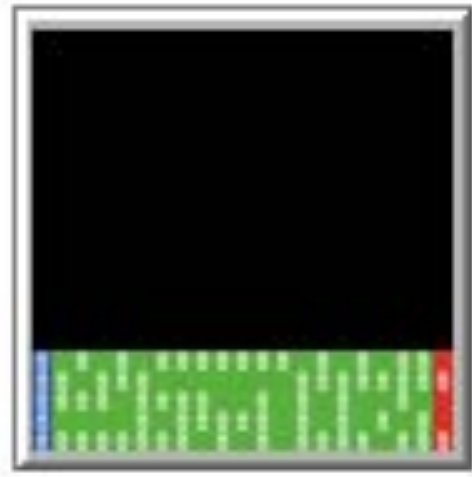
Figure 7.

This phenomenon can be extended to most turns, an extension that be-

comes apparent when examining corners. **Figure 8a** has outflow of 2.0 e/s, whereas the model represented by **Figure 8b** has outflow of 2.6 e/s, an increase of 30%.



a. Corner flow: flow to exit at upper right is 2 e/s.



b. Linear outflow: flow to exit at lower right is 2.6 e/s.

Figure 8.

We propose the presence of emergency personnel to direct evacuee traffic such that their flows more closely resemble that of **Figure 7b** rather than **Figure 7a** and that of **Figure 8b** rather than **Figure 8a**. Moreover, any future secret exits constructed should be oriented with this phenomenon in mind, i.e., facing the largest open corridor.

Part II: The Global Flow Model

This section abstracts the problem into a network flows formulation, solves for an optimal solution, and discusses model adaptability.

Abstraction as a Network

We use a graph to represent the sections and the connections among them. We represent our subsections as vertices and pathways between them (hallways or staircases) as edges. **Figure 9** shows the resultant graph representation. We denote nodes by two letters, such as L_A, for “lower ground A.” The rest of the nodes, not labeled with an L, correspond to the Napoleon Hall N, with G for “ground,” F for “first floor,” and S for “second floor.” For reference, the A–E lettering of sections is shown in **Figure 2**.

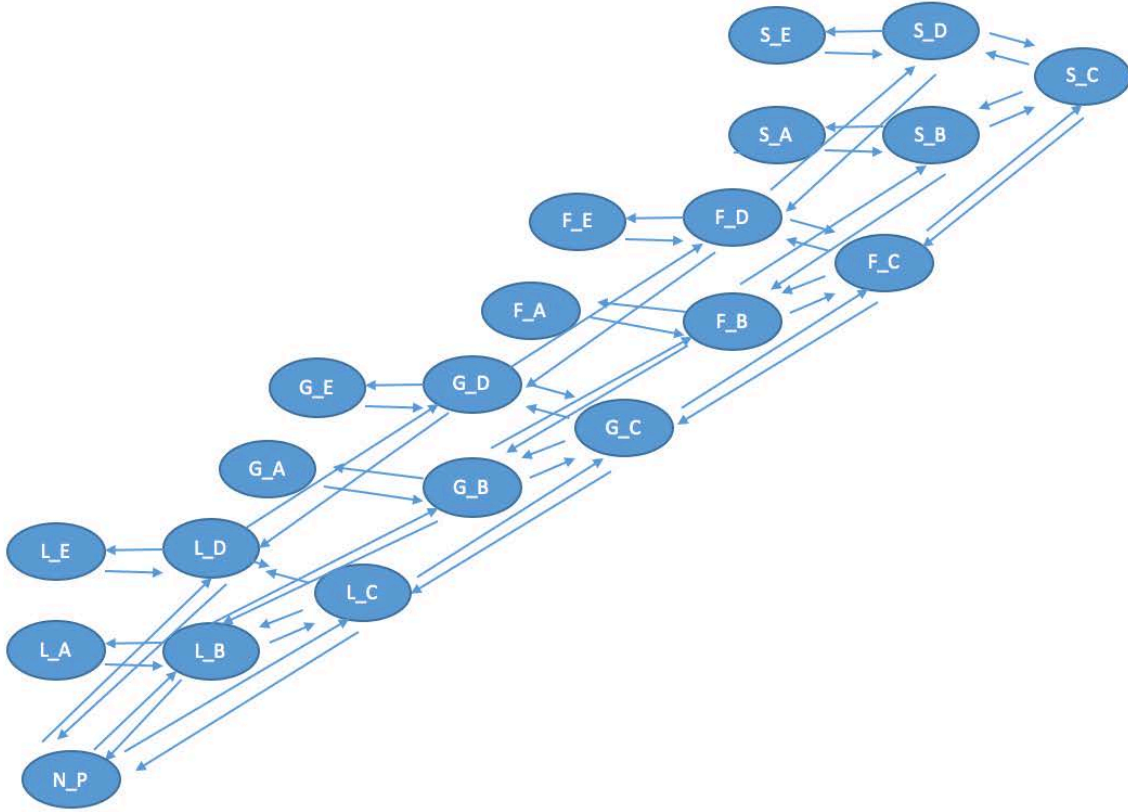


Figure 9. General graph showing all possible edges.

Network Flows Formulation

To formulate this problem in an abstract setting, we introduce the *network maximum flow* problem [Wikipedia 2019b]. Consider a directed graph $G = (V, E)$ with source vertex $s \in V$ and sink vertex $t \in V$. Additionally, define a capacity function c on the edges: For directed edge $e = (u, v)$, let $c(u, v) : E \rightarrow \mathbb{R}^+$. An s - t flow on a network is a function $f(u, v) : E \rightarrow \mathbb{R}^+$ that satisfies the following constraints:

$$\text{Skew symmetry: } f(u, v) = -f(v, u), \forall (u, v) \in E$$

$$\text{Capacity: } f(u, v) \leq c(u, v), \forall (u, v) \in E$$

$$\text{Balance: } \sum_{v \in V} f(u, v) = 0, \forall v \in V \setminus \{s, t\}$$

Notice how the balance constraint excludes s and t . This is because the value $|f|$ of a flow f is defined as

$$|f| = \sum_{v \in V} f(s, v) = \sum_{v \in V} f(v, t).$$

The maximum flow problem seeks the flow of maximum value on a network.

Intuitively, the flow problem can be thought of as sending water through a network of pipes as edges and vertices as junctions. Then

- The skew symmetry constraint says that the amount of water sent across an edge is equivalent to a negative amount sent in the opposite direction.
- The capacity constraints say that pipes have maximum amounts of water that can flow through them per unit time.
- The balance constraints say that each unit of water that flows through any non-terminal junction must also flow out.

The key question is, how much water can be sent from the source to the sink through the pipes in one unit of time? A specific example is shown in **Figure 9**.

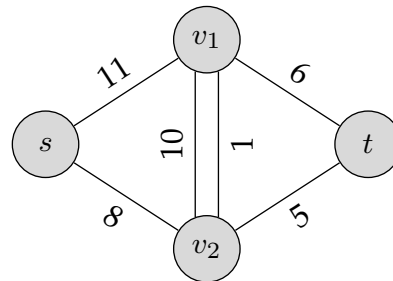


Figure 9. Network with capacities written on the edges. Maximum flow = 11 (send 6 on the top path and 5 on the bottom path).

To cast the evacuation problem as a network flow problem, we need to make slight modifications to the network structure described earlier. The classical network flow problem assumes a single source and a single sink, but evacuation of the Louvre would have visitors starting in multiple sections of the museum.

We create a “super-source” vertex s with edges to each subsection r of the museum. This extra set of edges can send at most the initial population of the room as a flow, so for subsection r we set $c(s, r)$ to be the initial population of the subsection.

Similarly, we add a “super-sink,” with a directed edge to the “super-sink” from every subsection from which people can exit the museum directly.

We use the Edmonds-Karp algorithm (1972) [Wikipedia 2019c] to find the maximum flow and the allocation of flows on the edges.

We interpret the maximum flow value m as the maximum number of people per second to leave the Louvre. This is an optimal rate under the constraints. If p people are in the Louvre when the evacuation is called, then the minimum time to evacuate the Louvre is

$$t_{\min} = \frac{p}{m}.$$

The value of $t_{\min}$ gives a heuristic to rank evacuation strategies, while also leaving room for discussion about how much it underestimates the actual time, particularly in the light of bottlenecks.

Finding Bottlenecks

To find bottlenecks, we introduce another abstract problem on graphs: the *minimum cut problem*. An s - t cut is defined by a subset C of edges such that removing C results in a disconnected graph in which s and t are in different components. The *cost* of a cut is given by

$$\text{cost} = \sum_{(u,v) \in C} c(u, v).$$

The minimum cut problem asks for a cut with minimum cost. The min-cut problem is intimately related to the max-flow problem by the *max-flow min-cut theorem* [Wikipedia 2019d]: The maximum flow from s to t in a network equals the cost of the minimum s - t cut.

Furthermore, since this theorem is a consequence of what is known as *strong duality* [Wikipedia 2019e] in linear programming, the optimal solution exhibits a property known as *complementary slackness*. This result says that the edges in the minimum cut are exactly the edges that have flow equal to their maximum capacity. In other words: *The edges that have maximum flow are the bottlenecks.*

Adapting our Model

Our model is flexible and can easily extend to the opening of any secret exits. The addition of a secret exit in a certain section can be represented by an edge from that vertex to the supersink, while blockages between sections can be represented by removal of corresponding edges. This technique of removing edges can also represent closing off additional passages, such as Passage Richelieu, by removing the corresponding edge to the supersink. This easy extension highlights the power of abstraction using graphs and network flows.

Louvre management can experiment with the parameters in the Net-Logo model; for example, they can observe the effects of increasing panic levels. Alternatively, they can set some agents to be significantly slower than others, to represent disabled evacuees.

Despite our concentration on the specifics of the Louvre's layout, our model is highly adaptable to other buildings, particularly since the Global Flow Model is not specific to any particular building and requires only accurate estimation of the edge weights.

Part III: Results and Recommendations

We discuss possible evacuation plans and results from our model. For our main model run, we allow as exits the entrances via the main Pyramid, the Passage Richelieu, the Carrousel du Louvre, and the Portes des Lions, with corresponding subsections the Passage Richelieu in lower ground D, Carrousel du Louvre in Ground E, and Portes des Lions in ground A.

Our evacuation plan calculates for each subsection the closest exit subsection, in terms of Euclidean distance, and sends all people toward that exit. We calculate a throughput to outside of 17.8 e/s.

The Louvre, with 8.1 million visitors in 2017 [Louvre Museum 2018] but closed on Tuesdays [Louvre Museum 2019] averages 26,000 people per day. Though the busiest days had significantly more visitors than average, it is unlikely that all the visitors were in the Louvre at the same time. So, we take 26,000 as a proxy for the worst case in an evacuation. Also, without information about the distribution of people, we simply allocate them uniformly across the rooms. With these assumptions, this evacuation plan takes

$$t_{\min} = \frac{26000}{17.8} \approx 1460 \text{ s} \approx 24 \text{ min.}$$

Bottlenecks

Figure 10 shows bottlenecks from each of L_D, G_A, and G_E to the supersink, as arrows on the left of the figure. These are the edges corresponding to the Passage Richelieu, Portes des Lions, and Carrousel du Louvre entrances. This means that the bottleneck for such a section is actually exiting the building, not getting to the entrance itself. This is in contrast to the N_P vertex for the Pyramid (leftmost arrow in the figure); for the Pyramid, the bottleneck involves getting to the Pyramid, not crossing through it.

The Pyramid is *Not* a Bottleneck

Napoleon P (the Pyramid) is the main entrance for the public. It is rate-limited by a set of stairs leading up toward ground level. As such, we would intuitively expect the edge leading from Napoleon P to the outside, the supersink, to be a part of the minimum cut and therefore a bottleneck that requires further analysis.

This turns out not to be the case. Examining the minimum cut, we see, by virtue of the edges from lower ground C to Napoleon P and from lower ground B to Napoleon P, that it is in fact entering the Pyramid that represents the true bottleneck, and not exiting the Pyramid itself. The stairs between L_B and N_P and between L_C and N_P are the bottlenecks for this particular evacuation plan.

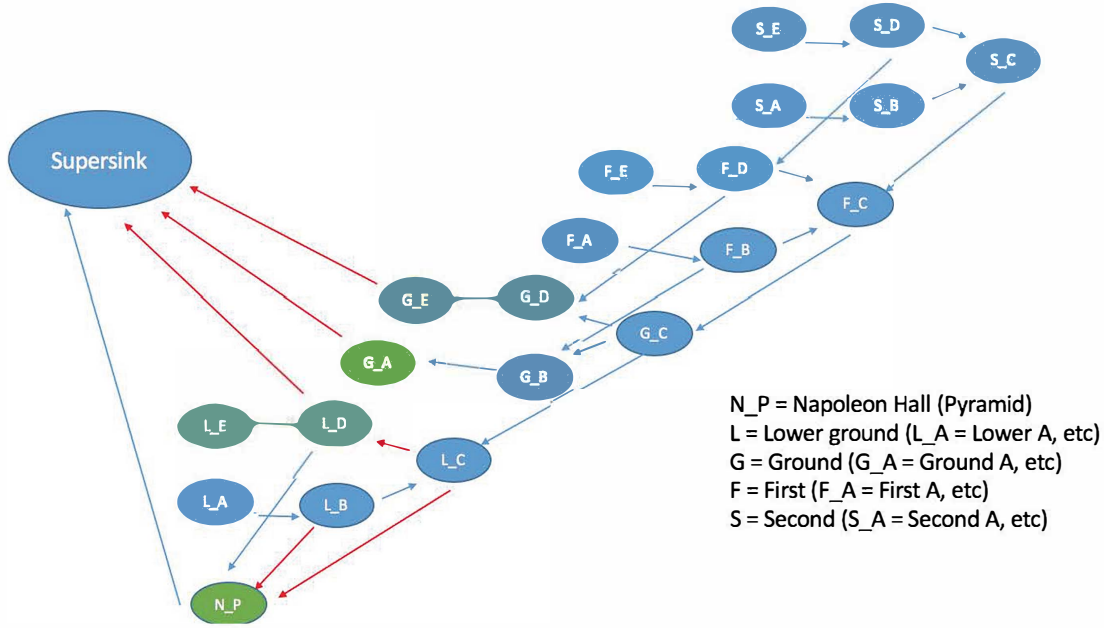


Figure 10. Bottlenecks, denoted by red arrows, from G_E, G_A, and L_D to Supersink, from L_C and L_E to L_D, and from L_C and L_B to N_P.

Our policy recommendation concerning the Pyramid is theoretically simple: Provide a higher access rate to it, via either new staircases or widening existing ones. A more feasible recommendation is to prioritize opening any secret exits surrounding the Pyramid, which may provide relief to the entryways into the Pyramid. More specifically, if secret exits exist in lower ground B or lower ground C, these exits would provide the most relief, given our current evacuation plan. We see in following sections why the Pyramid exit has such a large effect on the efficacy of any evacuation plan.

Passage Richelieu is Critical to Evacuation

The most important section in our evacuation plan is lower ground D, because of the relief it provides for entry into the Pyramid. We see this not only because Passage Richelieu is a part of our minimum cut, but also because removal of the edge (L_D, S) connecting lower ground D to the supersink constitutes a 6.8 e/s reduction in throughput. This would reduce our initial 17.8 e/s to a modest 11.0 e/s and our estimated total evacuation time from 24 min to 39 min, a staggering 64% jump in evacuation time. (In fact, (L_D, S) has a capacity of 7 e/s; the 0.2 e/s discrepancy comes in a rerouting in our max-flow network. This rerouting is an example of the adaptability of our model.)

More importantly, however, we observe that the removal of (L_D, S) changes our minimum cut dramatically; in particular, the edge from the Pyramid to the supersink, denoted (P, S) , becomes a bottleneck. For the graph, this is irrelevant; outflow is outflow, wherever it may come. Quali-

tatively, however, safety concerns around the Pyramid are more dire than around our other exit bottlenecks. The Pyramid's glass composition is one concern; another is that the Pyramid's status as Louvre icon and main public entrance makes it a target in the event of external attack. As a result, protecting the Passage Richelieu is also a form of protection for the Pyramid.

Our policy recommendation is therefore increased security for the Passage Richelieu. This matches well with the fact that the Passage Richelieu, at 7 e/s, has the largest exit throughput of any of the four public entrances. In addition, its additional utility in reducing strain on the Pyramid entrance is highly important to safe evacuation.

Ground D is a Bottleneck

The most restrictive edge in our minimum cut, and therefore the most powerful bottleneck on the outflow of evacuees, is the edge connecting ground D to ground E. In fact, this D–E edge actually allows a maximum throughput of only 1.8 e/s, an unexpectedly small number given the width of the corridor connecting ground D and ground E, modeled conservatively at 4 patches, or roughly 20 ft.

To understand this discrepancy, we turn to our Local Section Model for insight. Our evacuation plan asks all evacuees in ground D to move toward the Carrousel du Louvre exit in ground E. However, their progress is impeded by narrow corridors and complex turns of complex and nested galleries, which are detailed in **Figure 4**. As a result, evacuees asked to traverse ground D towards the corridor leading to ground E are blocked by other evacuees, leading to the bottleneck described above as well as to increasing levels of panic in the population. This situation is particularly problematic because ground D is the only section that leads to ground E. Note in **Figure 11** how many evacuees are trapped in the middle common area, and how correspondingly few evacuees are able to push through to the exit. Moreover, rising panic levels, overcrowding, and the resulting public safety dangers contribute to difficulties beyond a limitation on exit flow.

We investigate the quantitative effect of ground D bottlenecking on total global flow. Since ground D is the only section connecting to ground E, the exit flow from the Carrousel du Louvre exit is $\min(F_e, F_d)$, where F_e and F_d are the flow out from the Carrousel du Louvre and the flow from ground D to ground E, respectively. Experimental results give $F_e = 4.2$ e/s and $F_d = 1.8$ e/s, for a total exit flow out from the Carrousel du Louvre of 1.8 e/s. Hence, the layout of ground D is directly responsible for a net loss of 2.4 e/s. Removing the bottleneck could therefore improve global exit flow from 17.8 e/s to 20.2 e/s, improving total evacuation time from 24 min to 21 min. This is the largest potential improvement for our model.

One other experiment is replacement of the Carrousel du Louvre exit



Figure 11. Bottlenecking in ground D. Sources (stairs and other hallways) are marked in blue (e.g., vertical line at upper right), while the exit corridor (at upper left) is marked in red. The most prominent bottlenecking section (at top) is boxed in purple, while an unused path (at bottom) is boxed in orange. An example of a secondary (corner) bottleneck is circled in yellow.

with an equivalent-size exit in ground D, placed precisely where the largest number of evacuees begin to crowd (**Figure 12**). Call this new exit “passage ground D.” Placement of passage ground D results in an exit flow of 4.8 e/s, as opposed to the 1.8 e/s provided by the Carrousel du Louvre exit. This value matches well with the value provided by the linear flow out of the Porte des Lions exit described by **Figure 7b**. Since the Porte des Lions exit has been modeled to the same size as the Carrousel du Louvre exit, this indicates that passage ground D has reached some level of optimality in terms of its exit throughput. Such a passage ground D would result in a global outflow of 20.8 e/s, reducing total evacuation time to 21 min, a decrease of 15%.

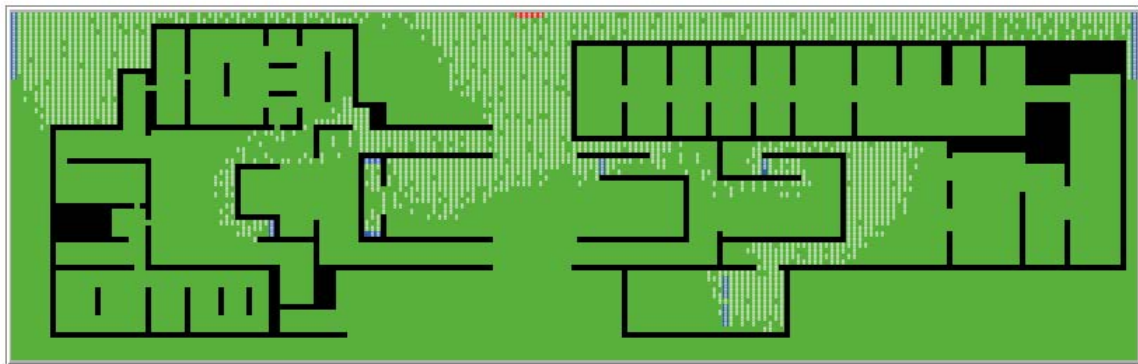


Figure 12. Placing an exit, marked in red (at top near center), where the largest number of evacuees begin to congregate, provides a network flow of 4.8 e/s as opposed to a base flow of 1.8 e/s.

Compared to its maximum capacity, the Carrousel du Louvre is underutilized in our evacuation plan—and in all evacuation plans that involve it, since ground E, where it is located, is connected only to ground D, so its throughput is limited by the large, complex network of galleries in ground D, especially by the section at the top left (outlined in purple) in **Figure 11**.

Consequent Recommendations

- The widening of the section **Figure 11** at upper left outlined in purple. The proximity of gallery walls to museum walls constricts the natural flow of evacuees towards the exit and causes increasing levels of panic as well. Widening would allow more evacuees to take advantage of this space.
- The placement of emergency personnel to direct evacuees around the “back,” through the section in **Figure 11** at lower left outlined in orange. Agents following their self-interest through the shortest path to the exit begin to squeeze towards the purple-outlined section, causing a bottleneck; by directing people through the orange-outlined path, a locally suboptimal path for one person would increase total throughput and therefore provide a globally more optimal solution. Alternatively, a software package such as a phone app to highlight unused or less crowded routes to a destination would also help to optimize the global evacuation situation.
- The removal of “secondary” and corner bottlenecks, such as that circled in yellow in **Figure 11**. Several of these exist in ground D, restricting evacuee flow and providing multiple bottlenecks to the exit point, trapping a large number of people in a small space and—in restricting their ability to feel progression towards their end goal—increasing the panic level. Some secondary bottlenecks may be hidden by the large flood of people that we use to extract maximal capacity from the exits. Providing simpler gallery structures would help produce a more natural flow of evacuees, potentially increasing throughput and lowering public safety concerns about overcrowding.
- More drastic recommendations center around the potential new exit depicted in red near the middle at the top of **Figure 12**. Its construction is beyond the scope of our recommendations; but if there is a secret exit located in ground D, its opening would provide the largest gain in exit throughput.

Braess’s Paradox

Surprisingly, removing edges from a network can actually increase traffic flow across it. This phenomenon is known as *Braess’s paradox*, which we illustrate with the example in **Figure 13**.

In **Figure 13**, assume that 4000 people are to cross from the start node to the end node. The time taken to traverse an edge is either 45 time units or $0.01P$ units, where P is the number of people using the edge.

First, consider the case where the edge labeled x does not exist. If A people take the v_1 path and B people take the v_2 path, then the times for the paths will be $0.01A + 45$ and $0.01B + 45$. Assuming that each person

takes the path to minimize their time, they will reach an equilibrium where each path takes the same amount of time, giving the following system of equations:

$$\begin{aligned} 0.01A + 45 &= 0.01B + 45, \\ A + B &= 4000. \end{aligned}$$

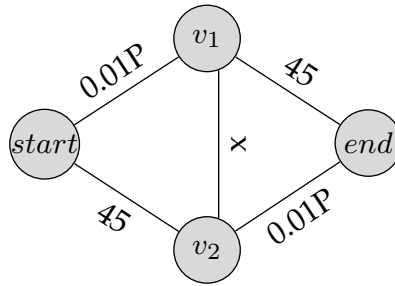


Figure 13. Example to illustrate Braess's paradox, from Valiant and Roughgarden [2010].

This system solves to $A = B = 2000$. This results in each person taking $45 + .01(2000) = 65$ units of time.

Now, consider the case where edge x exists but takes no time to traverse. For a single person who takes the path $start-v_1-v_2-end$, the total time becomes $0.01(2000) + (0.01)(2001) \approx 40$, a 25-unit saving. However, that saving is an incentive for more people to try this route; with each additional person, the time for the route increases, until 2500 people take the route, at which point it takes $0.01(2500) + 0.01(4000) = 65$ units of time, the same as before when there was no edge x . However, those taking only the v_2 path will find that their route now takes 85 units of time, so they too are incentivized to take the $start-v_1-v_2-end$ path. Then everyone's path takes $.01(4000) + .01(4000) = 80$ units of time, and anyone taking either of the original paths would require $45 + .01(4000) = 85$ units. The addition of an extra route has actually made the rate worse for everyone!

Valiant and Roughgarden [2010] show that with high probability, such a paradox will occur in a network.

Per this discussion, we recommend for any sections of the Louvre with a middle corridor (such as the B subsections that we defined) that emergency personnel be placed there to route the people out of the middle corridor. Not allowing anyone to take the middle corridor would allow a higher global evacuation flow.

A Residual Network Gives Paths for Emergency Personnel

Another consequence of the Global Flow Network is that by using the max-flow solution over our constructed graph, we can find a *residual network* to find entry pathways for emergency personnel [Blum 2015]. For

capacity function $c(u, v)$ and flow function $f(u, v)$, the residual capacity function is $c_f(u, v) = c(u, v) - f(u, v)$. The residual network G_f has the same edges and vertices as G but with the residual capacity function on its edges. Note that the directed edges in G may be reversed in direction in the residual network. A residual network is such that a path from any exit point (Pyramid, Carrousel du Louvre, Passage Richelieu, or Porte des Lions) to any interior section of the museum exists if and only if there exists a path from the supersink to the interior node in the residual graph. By using a pathway in the residual network, emergency personnel entering will not obstruct the outflow of evacuees. This highlights another reason for securing the Passage Richelieu; by keeping the edge from the supersink to the Pyramid out of the min-cut, i.e., making sure that it is not a bottleneck, we ensure a pathway for emergency personnel to enter the museum.

Hence we recommend:

- Prioritize evacuation plans such that there exists an entry pathway in the residual network; in many ways, this can be summarized by preventing the Pyramid exit from becoming a bottleneck.
- Use the residual network to find entry pathways for emergency personnel.

Summary of Policy Recommendations

- **Recommendation 1:** Increase access to the Pyramid.
- **Recommendation 2:** If the situation calls for opening secret exits, prioritize those in subsections surrounding the Pyramid, especially in ground D.
- **Recommendation 3:** Increase security presence around the Passage Richelieu and place high priority on securing its efficacy for evacuation use, in order to secure also the safety of the Pyramid exit.
- **Recommendation 4:** Use emergency personnel to direct evacuees
 - away from middle corridors and so that the flow towards exits resembles the linear flow detailed in **Figure 7b**.
 - around the “back” of ground D, i.e., the section highlighted in orange in **Figure 10**. Alternatively, implement technology such as a phone app to assign paths to individuals or groups.
- **Recommendation 5:** Design any future new secret or public exits for a linear flow through the nearest entryway or corridor.
- **Recommendation 6:** Remove corners and complications from sections such as ground D, where complex gallery layouts produce secondary bottlenecks. Simpler gallery layouts would increase evacuation efficacy dramatically.

- **Recommendation 7:** Prioritize evacuation plans for which the residual network has paths into the museum. Use such paths for emergency personnel to enter the museum.
- **Recommendation 8:** Use technology and appropriate signage to mitigate the effects of language barriers on a diverse population.

Evaluation of the Model

Sensitivity Analysis

Our Global Flow Model is quite robust; it always finds an optimal solution. The model displays sensitivity mostly in the Local Section Model, due to the high-variance nature of agent-based models and the need for several meaningful hyperparameters to describe human behavior.

As a result, our sensitivity analysis involves varying some of the parameters in the NetLogo implementation of the model. In particular, we focus on varying the following 4 parameters: `repulsion factor`, `speed-stdev`, `panic-increment`, and `ppp` (people per patch). We vary one at a time for each of three subsection types of gallery, tight corridor, and corner. The base values for the parameters are shown in **Table 1a**. For sake of comparison, we also provide in **Table 1b** the flow capacities found for the default parameter values.

Values for the non-varied parameters are: `repulsion factor` = 1, `speed-stdev` = 0.15, `panic-increment` = 0.2, `ppp` = 2.

Table 2 shows the flows for the three subsections for changing the values to `repulsion factor` = 1, `speed-stdev` = 0.15, `panic-increment` = 0.2, and `ppp` = 2, while holding the other parameters' values at their default values.

Table 1.

a. Default values of parameters. **b.** Flow capacities of section types for the default values.

Section Type	Default Constants	Section Type	Flow Capacity (e/s)
<code>repulsion factor</code>	0.35	Gallery	1.8
<code>speed-stdev</code>	0.0365	Tight corridor	2.6
<code>panic-increment</code>	0.1	Corner	2.0
<code>ppp</code>	1		

From the tables, we see that the model is sensitive to the values of the parameters in terms of raw output, as expected. However, the relative ordering of the three types of subsection (gallery, tight corridor, and corner) does not change across the experiments, which is what we would expect given our qualitative understanding of the complexity of each layout. Also, none of the results are unreasonable, which fact shows both stability and, importantly, meaningfulness in the choices of parameters. A poorly

Table 2.

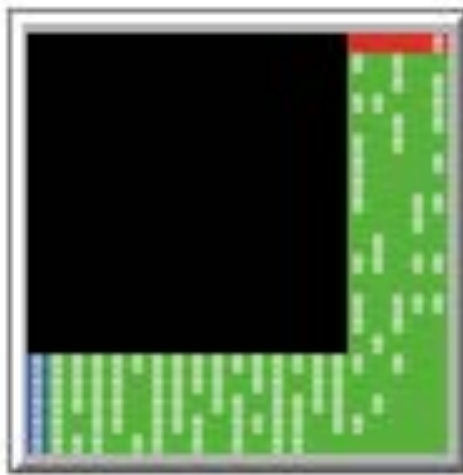
Flow capacities (e/s) for section types with varied values of some parameters, holding the other three parameters at default values.

Section Type	repulsion factor 1	speed-stdev 0.15	panic-increment 0.2	ppp 2
Gallery	1.1	1.0	1.1	2.4
Tight corridor	2.2	2.5	2.6	4.8
Corner	1.9	1.9	2.0	4.4

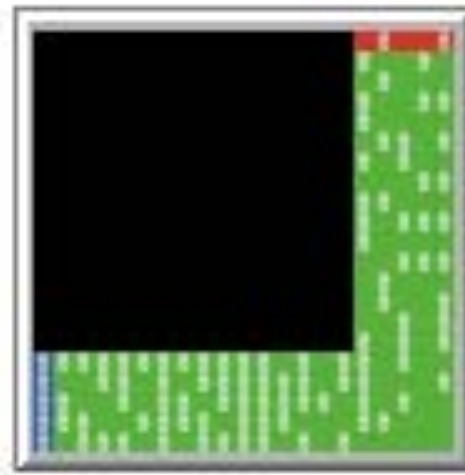
developed computational model with heavily tuned parameters would not be able to simply double or triple some of its parameters without resulting in unreasonable answers.

We comment individually on the parameters varied:

- Increasing repulsion factor increases agent aversion to crowded areas, pushing them away from each other. As a result, fewer people tend to cross the exit line. However, an interesting side effect of a high value of repulsion factor is that its effect is similar to the default parameters across a corner. Inspection of this phenomenon reveals that a high repulsion effect actually leads to more-optimal turning behavior, as shown in **Figure 14**. More specifically, high repulsion induces agents to use the width of the corner more effectively—a very rare instance of locally greedy behavior leading to globally optimizing behavior.



a. Effect of corners on exit flow: 2 e/s.



b. High-repulsion outflow, which also totals almost 2 e/s per despite the general negative effect of increased repulsion on exit flow.

Figure 14.

- The increase in standard deviation of walking speeds would, a priori, appear to have no effect because of the symmetry in the sampled normal distribution. However, this is not true, because the walking speed of a group is controlled by the slowest moving member in front. As a result,

having some people move even more slowly would cause the people behind them to slow down as well, so our results have reasonable real-world interpretations as well.

- We find that panic-increment heavily affects a gallery but neither a corner nor a tight corridor by much. The increase in panic comes into play only because of the complicated layout in the gallery. In the other two subsections, people can easily maintain movement, so panic does not increase nearly as much, which makes sense: Panic is most important if people are trapped in more complicated, more maze-like surroundings.
- For ppp, the observed increase in rate matches our expectation that more people per patch would indicate a large final exit throughput.

Overall, our sensitivity analysis shows that the model does not behave erratically for large increases in parameter values. Not only does this imply that the model is stable, but that the Louvre management will be able to test a large range of scenarios and plans and still retain the full utility of our model.

Strengths & Weaknesses

Strengths

- Agent-based modeling allows observing how complex behaviors emerge from a set of simple rules. In addition, the NetLogo implementation provides an intuitive interface for changing parameter values. We also took many steps to ensure physical interpretability so that we could map solutions back to the real world.
- Breaking up the Louvre into multiple simpler subsections allows us to observe and understand agent behavior within specific subsections and also behavior that leads to bottlenecks.
- Our Global Flow Model employs a high level of abstraction that makes it easy to integrate results from the Local Section Model to assess the entire system. Also, strong duality allows for a very natural interpretation in terms of global bottlenecks.
- The high level of abstraction naturally lends itself to adaptability. Test scenarios such as secret or blocked pathways can easily be mapped onto the Global Flow Model and incorporated into the strategy.
- The Local Section Model is quite robust in regard to large changes in the parameters. Additionally, the results of parameter changes have reasonable interpretations in the context of the situation.

Weaknesses

- Maximizing throughput, as opposed to minimizing time elapsed, yields an evacuation plan that is theoretically optimal for the worst case but may not be the best in average or more common cases.
- The Local Section Model inherently models both the Louvre layout and agent pathfinding in a discrete context. Given sufficient computational power, a more continuous model could give more realistic results.
- The agent logic in our Local Section Model places a very heavy emphasis on distance from an exit, which causes the majority of people to crowd toward the nearest exit. Only when huge queues of people form behind bottlenecks are some agents able to find alternative exits.
- There is a discrepancy between the local (selfish) optimization by agents in the Local Section Model and overall flow optimization of the Global Flow Model.

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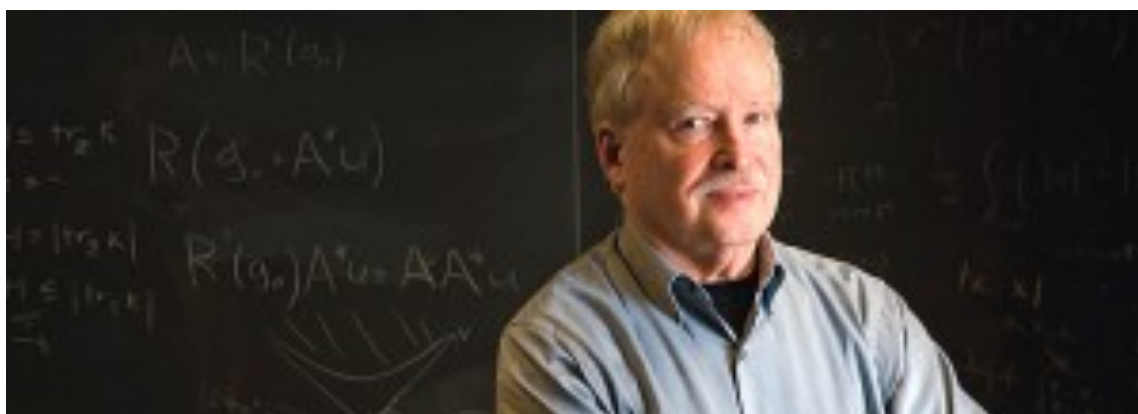
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